Table 1: Example 1 Monte Carlo results (PL parametrization). Entries are mean ± sample standard deviation over 30 runs. Timing is total wall-clock time; BI-HMC includes warm-up and sampling and is the same saved Python reference in both parametrizations. Skipped methods are omitted. BI-HMC sampler diagnostics flagged 30 of 30 fits; their finite point estimates are included.

Method	Parameter error (%)	Output RMSE	Wall time (s)
RLS	1.2484 ± 1.5001	0.019321 ± 0.015577	0.32638 ± 0.19494
WS-GNI	1.2484 ± 1.5001	0.019321 ± 0.015577	0.29916 ± 0.18481
WS-GGI	1.4922 ± 0.9506	0.019389 ± 0.015449	1.989 ± 0.426
GHAM-1-dH	1.2484 ± 1.5001	0.019321 ± 0.015577	1.297 ± 0.270
GHAM-2-dH	1.2484 ± 1.5001	0.019321 ± 0.015577	0.65640 ± 0.16140
GHAM-2-I	0.8055 ± 0.9286	0.019240 ± 0.015480	2.188 ± 0.484
LHAM-1-TIK	1.0519 ± 1.4131	0.019299 ± 0.015555	2.034 ± 0.760
LHAM-3-TIK	1.2397 ± 1.4493	0.019310 ± 0.015571	1.221 ± 0.742
BI-HMC	1.2513 ± 1.5074	0.019249 ± 0.015613	37.325 ± 6.649

Table 2: Example 1 Monte Carlo results (FSM parametrization). Entries are mean ± sample standard deviation over 30 runs. Timing is total wall-clock time; BI-HMC includes warm-up and sampling and is the same saved Python reference in both parametrizations. Skipped methods are omitted. BI-HMC sampler diagnostics flagged 30 of 30 fits; their finite point estimates are included.

Method	Parameter error (%)	Output RMSE	Wall time (s)
Newton	1.2494 ± 1.5023	0.019250 ± 0.015611	46.814 ± 6.537
GI	1.2494 ± 1.5023	0.019250 ± 0.015611	22.260 ± 2.493
GHAM-1-dH	1.2494 ± 1.5023	0.019250 ± 0.015611	46.563 ± 6.148
GHAM-2-dH	1.2492 ± 1.5024	0.019248 ± 0.015614	46.735 ± 5.888
GHAM-2-H	1.2492 ± 1.5024	0.019248 ± 0.015613	47.107 ± 4.760
GHAM-2-I	1.2492 ± 1.5024	0.019249 ± 0.015613	47.849 ± 4.863
LHAM-1-TIK	1.2494 ± 1.5023	0.019250 ± 0.015611	23.994 ± 2.326
LHAM-3-TIK	1.2494 ± 1.5023	0.019250 ± 0.015611	20.524 ± 2.298
BI-HMC	1.2513 ± 1.5074	0.019249 ± 0.015613	37.325 ± 6.649

Table 3: CSTR Monte Carlo results (PL parametrization). Entries are mean ± sample standard deviation over 30 runs. Timing is total wall-clock time; BI-HMC includes warm-up and sampling and is the same saved Python reference in both parametrizations. Skipped methods are omitted. BI-HMC sampler diagnostics flagged 30 of 30 fits; their finite point estimates are included.

Method	Parameter error (%)	Output RMSE	Wall time (s)
RLS	2.0868 ± 2.3168	0.014977 ± 0.011308	0.19974 ± 0.06690
WS-GNI	2.0868 ± 2.3168	0.014977 ± 0.011308	0.19481 ± 0.08445
WS-GGI	18.5020 ± 1.1788	0.026233 ± 0.008157	2.752 ± 1.130
GHAM-1-dH	2.0867 ± 2.3168	0.014977 ± 0.011308	2.651 ± 0.370
GHAM-2-dH	2.0868 ± 2.3168	0.014977 ± 0.011308	1.518 ± 0.274
GHAM-2-I	23.6905 ± 0.7712	0.030690 ± 0.007308	2.889 ± 0.468
LHAM-1-TIK	3.2239 ± 2.3158	0.015259 ± 0.011295	2.974 ± 0.453
LHAM-3-TIK	2.2958 ± 2.4054	0.015052 ± 0.011314	2.504 ± 0.653
BI-HMC	2.0538 ± 2.3272	0.014908 ± 0.011352	37.134 ± 6.039

Table 4: CSTR Monte Carlo results (FSM parametrization). Entries are mean ± sample standard deviation over 30 runs. Timing is total wall-clock time; BI-HMC includes warm-up and sampling and is the same saved Python reference in both parametrizations. Skipped methods are omitted. BI-HMC sampler diagnostics flagged 30 of 30 fits; their finite point estimates are included.

Method	Parameter error (%)	Output RMSE	Wall time (s)
Newton	2.2587 ± 2.2092	0.015451 ± 0.011096	48.103 ± 5.770
GI	2.2637 ± 2.2067	0.015467 ± 0.011089	22.322 ± 2.842
GHAM-1-dH	2.3162 ± 2.1809	0.015634 ± 0.011014	47.148 ± 5.710
GHAM-2-dH	3.5796 ± 1.6875	0.020899 ± 0.009236	47.430 ± 5.071
GHAM-2-H	3.3904 ± 1.7502	0.020043 ± 0.009485	47.549 ± 4.986
GHAM-2-I	3.4098 ± 1.7425	0.020136 ± 0.009458	47.518 ± 3.984
LHAM-1-TIK	2.2587 ± 2.2092	0.015451 ± 0.011095	16.993 ± 1.651
LHAM-3-TIK	2.2588 ± 2.2093	0.015451 ± 0.011096	13.711 ± 1.325
BI-HMC	2.0538 ± 2.3272	0.014908 ± 0.011352	37.134 ± 6.039